

CC261 - Computer Networks**Credit Hours:** 3(2+1) **Prerequisites:** None**Course Category:** Core Computing Course**Description**

This course introduces the basic concept of computer network to the students. The network functionalities, procedures, protocols and various terminologies are discussed by taking the layered network models (OSI, TCP/IP). A simulation tool is also discussed. In the practical work the packet tracer is used for building different types of networks.

Textbook(s)

- James F. Kurose, Keith Ross. Computer Networking 8th Edition, Pearson, (2020), ISBN: 978-0136681557
- William Stallings. Data and Computer Communications, 10th Edition, Pearson, (2013), ISBN: 978-0133506488
- Behrouz A. Forouzan. Data Communication and Computer Networks, 5th Edition, McGraw Hill, (2012), ISBN: 978-0073376226

Other supplemental materials

- Packet Tracer Tutorial -
<https://www.packettracernetwork.com/tutorials/>

- Online Networking Tools -
<https://manytools.org/network/>

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
1	CLO Describe the key terminologies and technologies of computer networks	C2 1,2,10
2	CLO Explain the services and functions provided by each layer in the Internet protocol stack.	C2 1,2,10
3	CLO Analyze various internetworking devices and protocols and their functions.	C4 4,5
4	CLO Build Computer Network on various Topologies	P3 4,5,6 ,8

Week wise Lecture Schedule

W #	Lecture Topics
1	Basic concepts of networking, Peer to peer and client server architectures,
2	Network topologies, LAN, MAN and WAN
3	Layered architecture, TCP/IP, OSI
4	Physical layer functionality, Broadcast vs. Point-to-Point Communication, Directional and Omni Directional Communication, Modes of Communication (Simplex, Half Duplex and Full Duplex).

W #	Lecture Topics
5	Media: Bounded Media, Coaxial Cable, Twisted Pair (Shielded and Unshielded), Optical Fiber, Unbounded Media, Radio Waves, Microwave, Infrared, Satellites
6	Data link layer functionality, MAC addressing (unicast, multicast, broadcast) , CSMA, Types of Transmission (Asynchronous and Synchronous Transmission)
7	Multiplexing: Frequency Division Multiplexing, Time Division Multiplexing, Synchronous TDM & Statistical TDM
8	IEEE 802.x standards (Wired Networks)
9	Networking devices, Network layer protocols (IPv4 and IPv6)
10	Transport layer protocols, ports and sockets, TCP and UDP
11	Application layer protocols
12	Wireless Networks standards and technologies
13	Decentralized wireless networks
14	Cellular Networks, Generations and advancements
15	Latest trends in computer networks
16	Case studies